GEK-85740

GE Drive Systems

INSTRUCTIONS

"OITCON"

OPERATOR INTERFACE CONSOLE

REMOTE ECMS TERMINAL

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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NOTES:

1. SAFETY PRECAUTIONS

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.
WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.
CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.
NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when installing or operating this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT DUE TO REMOTE ENERGIZATION OR OPERATION OF EQUIPMENT HAZARD: No modification to the equipment or software should be attempted without first reviewing the effect on the overall system and taking the necessary precautions to assure safety of personnel.

WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT DUE TO REMOTE ENERGIZATION OR OPERATION OF EQUIPMENT HAZARD: Assure safety of personnel throughout the entire system before operating this and associated equipment.

WARNING - ELECTRICAL SHOCK AND ROTATING EQUIPMENT INJURY HAZARD: Read and observe all safety precautions listed in the drive controller Instruction Book (GEK) when operating or troubleshooting this equipment.

CAUTION: Do not remove connections or printed circuit cards while power is applied. This can damage the equipment.

CAUTION: Installation wiring must be in accordance with the National Electrical Code and be consistent with all local codes.

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow the start-up procedures step by step.*

NOTE: *Read and heed all **WARNING, CAUTION** and **NOTE** labels posted on the equipment.*

2. RECEIVING AND STORAGE GUIDELINES

RECEIVING

The equipment and software should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVE SYSTEMS, ERIE, PA.** When seeking assistance please use **serial number, requisition number, model number** and **drive code** to identify the equipment **(Telephone 814-875-3851)**.

STORAGE

Hardware:

If this equipment is not to be installed immediately, hardware should be stored in a clean, dry location at ambient temperatures from -20° C (-4° F) to 55° C (131° F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage enviroment exceeds a 15° C (27° F) drop in temperature at 50% humidity over a four hour period, a heat source should be provided to prevent condensation. (A well placed light bulb can sometimes serve as a source of heat). Higher humidities with smaller temperature changes can also cause condensation.

Software:

Software provided on magnetic media, such as floppy disks and magnetic tape, should be stored in a clean, dry location at ambient temperatures of 10° C to 55° C (50° F to 125° F). Care should be taken to protect the media from magnetic fields, high heat or humidity and physical damage.

IMPORTANT: *If problems are encountered during the installation or operation of this terminal, contact GE Drive Systems (814-875-3851) with specific details of the problem.*

3. INTRODUCTION

The OITCON (Operator Interface Console) is a remote terminal that can be connected to a GE Extended Crane Management System (ECMS) via a serial link (RS422). The ECMS computer runs a multitasking, multiuser operating system called 386 CONCURRENT DOS. This operating system allows up to **six** independent consoles (or operators) to work from the main processor. **Four** of these consoles are located on the MAIN ECMS SCREEN, allowing **two** more to be located remotely from the ECMS computer. The OITCON is an intelligent terminal configured as one of these remote consoles.

The ECMS monitors crane operation by examination of the Series Six™ Programmable Controller's memory. Thus the ECMS screen can show the operation of the crane (devices being turned ON and OFF), internal operation, faults in the GE drives (drive information is transferred to the Series Six by the local area network), and operation and faults from the GE Genius™ I/O system (connected by the Genius Bus to the Series Six Programmable Controller). The OITCON has direct access to the same information through the ECMS.

4. PRINCIPLES OF OPERATION

The ECMS computer runs a multi-tasking, multi-user operating system. The OITCON is connected into the ECMS computer via an RS422 serial link operating at 19,600 baud, as one of the two possible remote terminals of this multi-tasking system. OITCON is configured as console # 5 on the ECMS. Since OITCON is a serial terminal, it cannot show screens composed entirely of graphic information (such as the trendgraphs), nor can it run programs contained on the ECMS computer that are specifically designed to address the main system console (the main screen on the ECMS itself). There are **two programs** of the ECMS computer located on drive C: and in subdirectory "MMI\EXECUTE" called OITCON.CMD and OITLOAD.CMD. OITCON.CMD is the main program designed to link the remote terminal into the ECMS database, and to display the ECMS screens on that terminal. It also allows access into the alarm and event section of ECMS so that information may be displayed remotely. OITLOAD.CMD is a program designed to convert the ECMS screen format files (named FOR##.IBM in the C:\MMI\PROJECT subdirectory) into OITCON formatted screen files (named FOR##.OIT). OITLOAD.CMD will also download selected screen files into the OIT memory for faster screen update.

NOTE: *As the OITCON is a remote device from the ECMS, powering up the OIT after the ECMS is running, or powering up the OIT during ECMS powerup, may cause the OITCON to operate incorrectly. Refer to the section on OITCON Recovery Proceedures in such an event.*

5. COMMUNICATING WITH ECMS

When power is applied to the OITCON, it performs a "self-test" of its internal systems, then assumes the "on-line" mode. The OITCON must be configured correctly for serial link speed, word length, stop bits, wrap, line feed, status lines, and type of link. These settings and how to check them are detailed in APPENDIX A "OITCON Configuration". If the settings are correct the OITCON will automatically connect to the ECMS via the serial link and become a remote terminal of the ECMS.

™ - Series Six is a Trademark of GE Fanuc
™ - Genius is a Trademark of General Electric Company.

Normally, the OITCON and the ECMS computer will be powered up together i.e., AC power to the OITCON will be applied at the same time as the AC power to the ECMS. This will ensure that the OITCON passes through the "start-up procedure" dictated by the ECMS computer at the correct time. When the OITCON has connected to the ECMS operating system, the following message will appear on the screen:

Concurrent DOS 386 3.0
Copyright (C) 1982,1988 Digital Research

Concurrent DOS 386 3.0
General Electric ECMS Version

C:\MMI\PROJECT>

The OITCON is forced to run the OITCON program on the ECMS, and after a few seconds, will display the ECMS LOGO SCREEN.

As indicated in the "Principles of Operation" section of this manual, if the OITCON is turned on at a different time than the ECMS computer, the normal start-up sequence will be interrupted. This may occur naturally sometimes as the ECMS is supported by an uninterruptible power supply (UPS), that will sustain power to the ECMS for 2 to 5 minutes after normal power is removed, to allow the ECMS to monitor the shutting down of the crane, and to have an ordered power down of the computer system. If power is restored to the ECMS within this time period, the ECMS will not shutdown, but continue operation. However, during this time, power to the OITCON may have been shut off. If power to the OITCON is restored before the end of the time period, the OITCON will proceed through its "SELF-TEST" and connect to the ECMS computer (nothing will be displayed on the screen). In this case, the operator should press function key **F10** on the OITCON to redisplay the ECMS LOGO SCREEN, and continue from that point.

If the OITCON start-up has been disoriented by unsequenced power-up, sometimes pressing F10 will not clear the problem. In this case, the operator should press the ENTER key on the OITCON, then press the "START OITCON" function key on the ECMS keyboard in the machine house.

If both above steps fail to start the OITCON, perform the following:

Switch off power to both the OITCON and the ECMS.
Wait until the UPS turns off the ECMS completely, then restore power to both the OITCON and the ECMS simultaneously.

6. MONITORING SCREENS

Once the ECMS LOGO SCREEN is displayed on the OITCON, the operator will have to choose which monitoring screen is activated by pushing the appropriate function key (indicated along the top of the screen).

Function key F1 will bring up the main "SYSTEM SCREEN" from the ECMS. This screen shows all the main functions of the electrical system on the crane -- AC Supply, DC Control, AC Motor Control, and Operator Inputs. This screen is designed to assist operators in diagnosing problems they may have in operating the crane.

Function key F2 will bring up the "OPERATING SCREEN". This screen displays relevent information needed in the daily use of the crane.

With either the "OPERATING SCREEN" or the "SYSTEM SCREEN" displayed, the operator has access to individual drive screens for more detailed monitoring and diagnostics.

Function key F2 (with either the OPERATING SCREEN or SYSTEM SCREEN displayed), for example, will call up the "HOIST SCREEN". This screen has been designed to show all the functions associated with the HOIST motion, and can be extremely helpful in locating the cause of a malfunction in this drive. On this display, the sequence of the drive flows from top to bottom -- from the **Undervoltage relay**, to the **Stop relay**, to operator requests, to events within the drive, to limits that may be affecting the movement of the machine. By following this natural sequence, the operator can determine the problem area.

Function key F6 will bring up the ALARM and EVENT CONSOLE from the ECMS. This function is described in the next section.

NOTE: *Pressing function key **F10** while viewing any of the MONITORING SCREENS will return the operator to the ECMS LOGO SCREEN and allow selection between the OPERATE and SYSTEM SCREENS.*

7. EVENT CONSOLE OPERATION

The ECMS will keep a running log of the last 2000 alarms and events that occurred on the crane. When the system was configured for the crane, some contacts changing state (e.g., from open to closed) were configured as "EVENTS" to be logged, some were configured as "ALARMS" to be logged, and some were for indication on the monitoring screens only. Those contacts configured as "EVENTS" will appear in the EVENT LOG Those contacts configured as "ALARMS" will appear in the ALARM LOG and the EVENT LOG.

This "log" is a circular that makes an endless loop 2000 units long. As each alarm and event occurs, it is added to the list until two thousand have been added. At this point, the first entry is deleted to make room for the 2001 entry, and so it continues. The list therefore contains the last 2000 alarms and events that occurred on the crane. Alarms and events occuring before that are gone.

Alarms and events can occur for a variety of reasons in the system during normal operation. All of these alarms and events are translated into a message TEXT that is sent to the alarm console program. The alarm console program provides a variety of user commands using function keys to display these TEXT messages.

ACTIVE ALARM DISPLAY

ACTIVE ALARMS are those messages that have either a priority of URGENT or CAUTION. When an URGENT or CAUTION ALARM ("high priority" alarms) occurs, it is always shown on the ACTIVE ALARM display. When it returns to normal, it is removed from the ACTIVE ALARM display.

The event console can retain up to 1000 "high priority" alarms for this display. When more than 1000 high priority alarms exist, the **oldest** alarms are removed from the list.

The top of the display screen shows the available functions to the operator. Each function is selected by pressing the appropriate function key. The bottom of the display shows the current event console display at the left side and the top event number on the right side.

The top alarm is a number representing a pointer into the list of 1000 high priority alarms. As you scroll through the list, the top alarm number changes to indicate where you are in the list.

Several keys are available to look at the active alarms. The PGUP (**F11**) and PGDN (**F12**) keys cause the display to page through **one screen** at a time. The CARRIAGE RETURN key scrolls up **one line** at a time. The UP and DOWN CURSOR keys allow moving up and down **one line** at a time. The HOME key brings you back to the top alarm.

FUNCTION KEY OPERATIONS

The operation of each function key is described as follows:

Active Alarms
This function will change the display to the ACTIVE ALARM display.

Event Subset
This function will display all alarms and events that fulfil the search conditions specified in the "LOAD SUBSET"" system.

Previous Page
This function will cause the screen to display the previous 22 active alarms. This function is **inactive** when the display is already at the top alarm number 1.

Next Page
This function will cause the screen to display the next 22 active alarms. This function is **inactive** when the display is already at the last alarm (#100).

Latest Alarms
This function brings the first display back up and sets the TAB bar to the latest alarm on the list. This function key should **always** be used first when looking at the event console.

Graph Event
This function is not used by OITCON.

Return to System
This function causes the OITCON to return to the previous monitoring screen.

More
This function key will access the next level of functions available. **They appear below.**

Load Subset
This function is used to interrogate the event history for specific alarm/event occurences. (See Subset functions later).

Alarm Totals
This function will total all occurences of all alarms contained in the history files.

Print Events
This function is not used by OITCON.

Help
This function provides a help screen for the alarm/event highlighted by the TAB bar.

Acknowledge Alarms
This function is not supported by OITCON.

Subset Functions
These functions allow the user to select a subset of the total history file to be displayed. The functions vary depending on the "tabbed" alarm highlighted when the "load subset" function key is pressed. The subset chosen can be a function of alarm type, and name (use question marks ? for a wildcard symbol in a name mask).

EVENT HISTORY DISPLAY

This display is used to show all of the system EVENTS and ALARMS including return to normal messages. The console can display the 200 most recent alarms and events and also records the last 1000 messages to a disk file on the ECMS. When more than 200 messages exist, the oldest messages are removed from memory. When more than 1000 messages have been stored to disk, the oldest are discarded.

The top of the display screen shows the available functions to the operator. Each function is selected by pressing the appropriate function key. The bottom of the display shows the current alarm console display at the left side and the bottom alarm number on the right side.

The bottom alarm is a number representing a pointer into the list of 200 alarms and events. As you scroll through the list, the bottom alarm number changes to indicate where you are in the list.

Several keys are available to look at the event history. The PGUP (F11) and PGDN (F12) keys cause the display to page through **one screen** at a time. The CARRIAGE RETURN key scrolls up **one line** at a time. The UP and DOWN CURSOR keys allow moving up and down **one line** at a time. The HOME key brings you back to the bottom alarm.

The function keys are described below:
While most of these functions have been previously described, several new ones are provided in this menu. These are described in the following paragraphs.

Active Alarms
This function changes the display back to the ACTIVE ALARM display.

Latest Events
This function sets the bottom alarm back to 200 so that the 22 most recent events are shown on the screen.

ALARM MESSAGE FORMAT(S)

This section is provided to help the user understand the information that is contained in each of the various types of alarm and event messages. Each category of alarm or event in the system has a fixed format. This section allows you to identify what type of message format is being used and how to decode the various fields in the message.

Each alarm begins with the time and date that the alarm message text was generated.

The next two to four characters define the general type of alarm that is being shown. The possible codes are listed below:

Type	Code	Description
1	OK	Return to normal message
2	DA	Digital Alarm
3	DK	Digital Acknowledge (reserved for future use)
4	HI	High Alarm
5	HK	High Alarm Acknowledge (reserved for future use)
6	LO	Low Alarm
7	LK	Low Alarm Acknowledge (reserved for future use)
8	MDCH	Drive Mode Change Alarm
9	SPCH	Setpoint change message
10	GENI	Genius Alarm
11	##	Drive fault or ECMS Program alarm

Type 11 ALARMS use a two digit number in conjunction with the text string following it to indicate what drive or ECMS program has generated the fault message. Alarm **types 1, 8, 9,** and **10** are considered EVENTS and are not displayed on the ACTIVE ALARM summary display. This is due to the fact that no return to normal condition for these alarms is available.

The following paragraphs describe each alarm category.

Digital Alarms 15:20:38 03Feb88 DA U 272 COMFAIL COMMUNICATIONS STALLD

The above alarm format is used for digital alarms. Digital alarms are used for indicating faults when one state indicates normal status and the other is a fault condition. The alarm priority is indicated by **the 'U'** character*. In this example, the priority is URGENT' The number of the alarm is 272. This number is internal to the ECMS and has no real user value. The 'COMFAIL' text is a 1 to 9 character nickname (tagname) that is unique for every alarm point in the system. It is followed by a description of the alarm point and its new state. In this case, the state is 'STALLD'. When the alarm returns to normal, the state will indicate 'HEALTY'.

Analog Alarms

Analog alarms are similar to digital alarms except that they provide the current value of the point when the alarm text message was created. In addition, the engineering units for the point are shown immediately following the current value.

Genius I/O Alarms

Genius I/O alarms are always indicated by the word 'GENIUS' in the description field. Genius alarms can indicate faults for bus controller(s), blocks, and points.

Drive Alarms 15:27:00 03Feb88 4 U HOIST LOSS OF AN AC PHASE F22

The drive drop number on the LAN is indicated by the number 4. The description of the alarm includes an english description of the drive along with an expanded fault description. The fault number is shown at the end of the message.

Drive Mode Change Alarms

This function is not supported.

Setpoint Change Messages

15:25:06 03Feb88 SPCH PA01SET PARISON POINT #01 0.0000 to 56.00 %
15:26:01 03Feb88 SPCH E1MXTSP EXT 1 M-PIPE MELT 0.0000 to 230.DEG

Whenever a point is changed by the operator using the CHANGE POINT function key, a message alarm is generated. This message records the setpoint change showing the nickname (tagname), its description, the old value, and the new value with its engineering units (if applicable).

ECMS Program Alarms

15:23:49 03Feb88 72 E SC-SETUP Execution from DISK complete F93
15:27:00 03Feb88 75 E RECORDER Pen channel started and sampling F203

Several of the programs in the ECMS system generate ALARM or EVENT messages to indicate system conditions. For example, when a new pen is connected in the recorder system, an EVENT message is generated indicating that a pen channel has started. When recipe operation is complete, an EVENT message is generated to indicate the request has been processed.

NOTE: *By "TABBING" to a particular ALARM or EVENT (by using the arrow keys), and pressing function key F10, on-line help will be offered by the system, describing the cause, location, and possible solutions for the specific alarm/event.*

**E is EVENT, C is CAUTION, and A is ATTENTION.*

8. CHART RECORDER SET-UP

On some occasions, it may be necessary to start the ECMS chart recorder logging data for maintenance and diagnostics purposes. The chart recorder function, as implimented on OITCON, is a subset of the main ECMS chart recorder function, and operates through the ECMS main console. In other words, the OITCON chart recorder function will interrupt processing on the main ECMS terminal in the electrical house, and force it to set-up the chart recorder. Therefore, this function should only be used when it will not affect another person working on the main console.

The OITCON chart recorder contains some "canned" set-ups than can be recalled to the screen and set in motion. The function will not allow the user to individually select signals to be recorded (as can be done on the main console). A description of the signals that are included in the canned set-ups can be retrieved by pressing the "LIST TAGNAME" function key.

NOTE: *Due to the reduced functionality of a serial terminal, chart recorder graphs cannot be displayed on the OITCON. To view graphical data, the main ECMS console must be used.*

9. MONITORING THE SERIES SIX PROGRAMMABLE CONTROLLER

It is possible to monitor individual Series Six Programmable Controller I/O and registers using OITCON. These screens are retrieved by pressing the "MORE" function key **F8** while displaying a MONITORING SCREEN. This function is specifically for maintenance personnel only. The user must enter a valid Series Six address into the entry fields on the screens to obtain on-line data. Entry fields are accessed by pressing the DOWN ARROW KEY on the OITCON.

10. RECOVERY PROCEDURES

1. If the OITCON screen display should become corrupted, pressing the **F10** function key will normally clear the screen and return the OITCON to the ECMS LOGO SCREEN.

2. If the OITCON start-up has been disoriented by unsequenced power-up, sometimes pressing **F10** will not clear the problem. In this case, press the ENTER key on the OITCON, then press the "START OITCON" function key on the ECMS keyboard.

3. If both above steps fail to start the OITCON:
   -- Switch power off to both the OITCON and the ECMS.
   -- Wait until the UPS turns off the ECMS completely, then restore power to both the OITCON and the ECMS simultaneously.

4. If the OITCON appears to be working, and function key descriptions appear at the top of the screen, but the rest of the screen is blank, it will be necessary to reload the OITCON memory from the ECMS computer. This can be done by repowering the OITCON, then switching the ECMS to CONSOLE 4 by holding down the CONTROL key (CTRL) while pressing the NUMBERPAD KEY 4 on the ECMS keyboard. Then type 'RELOAD' on the ECMS keyboard. (This will run the batch file RELOAD in CONSOLE 4 that will force the program OITLOAD to run from CONSOLE 5, the OITCON, and download the required screens to the OITCON).

APPENDIX A: OITCON CONFIGURATION

(For Information Only)

In order to configure an OITCON, the following steps must be performed: (**NOTE**: A remote terminal will not operate on the ECMS unless the operating system set-up has been enabled.)

1. Configure the Operator Interface Terminal as follows:
   a. With a keyboard attached to the OIT, turn on power.
   b. Once OIT has completed self test, press Ctrl 1 to call up the set-up system.
   c. The OIT should be configured to the following parameters:

Component/Function	Configuration
Mode of Operation	Online ANSI No Echo Send any case
Date/Time	
End Line / Color	No Wrap No Auto LF Color
Cursor/Screen	Blinking Underline CRT Saver Off
Status Lines	One
Keyboard Codes	Five
Serial Port 1 (Primary)	19.2k None 8 Bit 1 Stop Soft Pt to Pt
Serial Port 2 (Secondary)	not used
Serial Port 3 (Tertiary)	not used
Configuration	Save
Confidence Tests	Diagnostic
Extended Memory	Save Screens 00 to Screens 00

NOTE: *Do not forget to save the configuration by highlighting the "SAVE" option and pressing* ***F6.***

2. Translate the ECMS Screens per the following guidelines:
   -- OITCON uses a different screen format than the ECMS.
   -- The OIT can only display 8 colors (including black and white) while the ECMS can display 16, therefore a translation of colors is also required.
   -- This translation process is performed by the OITLOAD.CMD program. Before OITLOAD can be run, one must decide which screens will be available to the OITCON.
   -- Screens in the ECMS system are identified by a number e.g., SYSTEM SCREEN = 30, HOIST SCREEN = 32, etc.
   -- Not all ECMS screens will be required on the OITCON.

3. Decide which (if any) of the screens for the OITCON, should be retained in the OITCON memory.
   -- Screens retained in memory will display faster when called up than screens that are kept on the ECMS hard disk.
   -- The OITCON can only retain between 6-8 screens in memory, therefore select only the most commonly used screens for memory storage.

4. Select desired screeens and construct OITLOAD command as in the following example:

   C:\MMI\PROJECT> OITLOAD 1 25f 30f 32f 34f 36f

   This example would load the following screens: 25 - OPERATE, 30 - SYSTEM, 32 - HOIST, 34 - TROLLEY, and 36 - GANTRY.

   NOTE: *The "f" after the screen number tags the screen for memory storage. If "f" screens are specified, the command must be issued from the OITCON itself while connected to the ECMS, and running as a remote terminal of the operating system.*

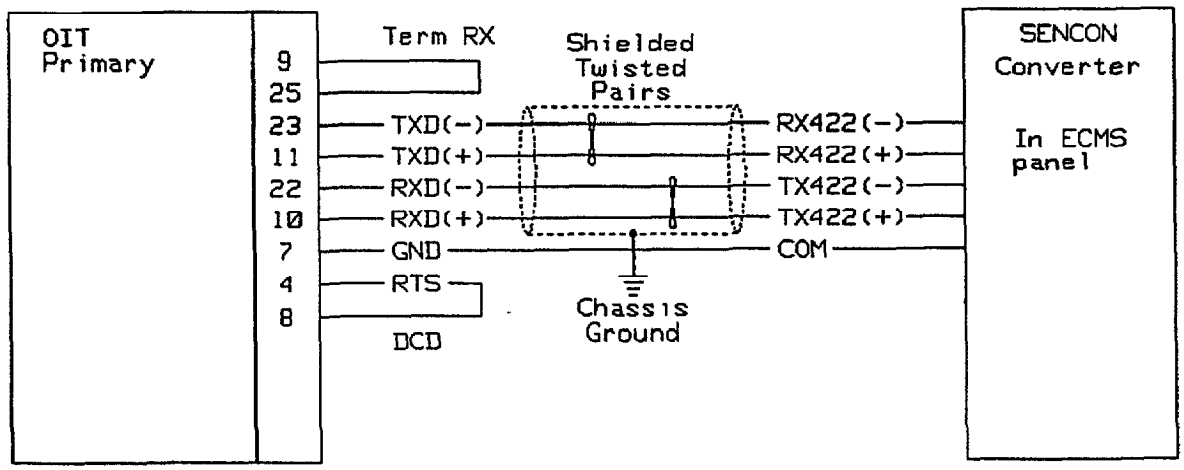
5. Create OITCON Function Keys using the following guidelines:
   - In the ECMS, function key assignments for the screens are contained in two files, FUNCTION.IBM and SPECIALS.IBM.
   - OITCON requires similar files called FUNCTION.OIT and SPECIALS.OIT.
   - Creating these two files is done by copying the two IBM files to OIT files. (Not all the function keys used on the ECMS screens are required for the OITCON.)
   - Perform the following in order to create "special set" of function keys for the OITCON.
     a. Copy FUNCTION.IBM to FUNCTION and SPECIALS.IBM to SPECIALS
     b. Edit the function key assignments using the ECMS configuration tool "CONFIG" Mto remove the unwanted function keys. (Remember the MORE levels as well).
     c. Exit the ECMS "CONFIG" system, then copy FUNCTION.IBM to FUNCTION.OIT and SPECIALS.IBM to SPECIALS.OIT.
     d. Rename FUNCTION to FUNCTION.IBM and SPECIALS to SPECIALS.IBM

The set-up of the OITCON is now complete. Note again, that the OITCON will not be recognized by the ECMS system unless enabled by GE.

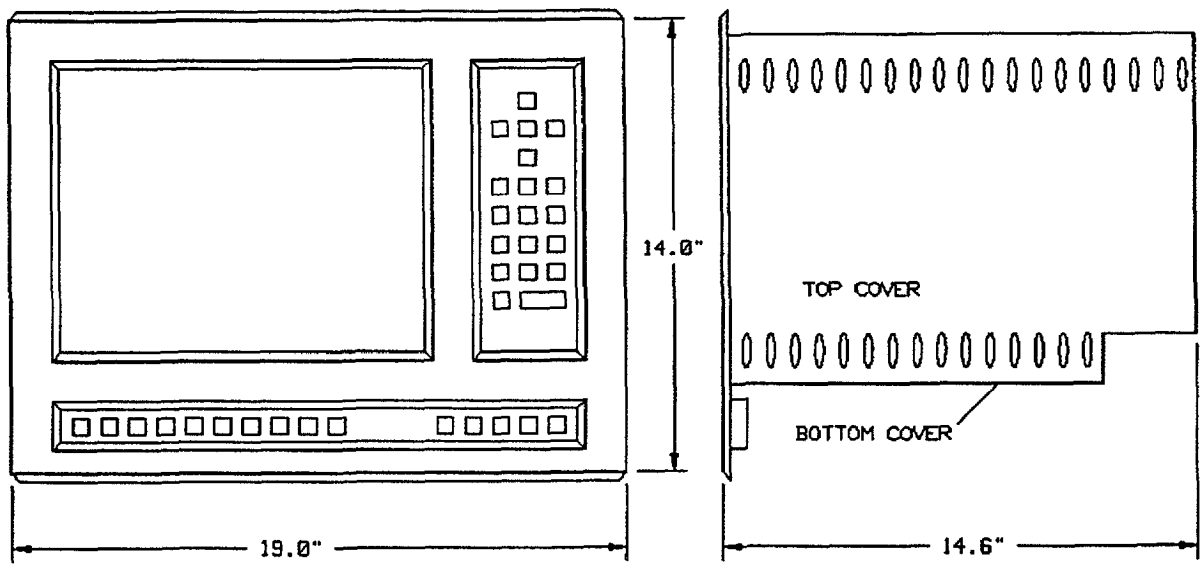
APPENDIX B: WIRING DIAGRAM AND OUTLINE

OITCON WIRING DIAGRAM

The diagram for cable connection between the OITCON and the ECMS computer is shown:



OITCON OUTLINE DRAWING





GE Drive Systems

GEK-85740 (2/92)

General Electric Company
GE Drive Systems
1100 Lawrence Parkway
Erie, PA 16531